Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

COMPUTER SCIENCE **9618/13**

Paper 1 Theory Fundamentals **October/November 2025**

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **16** pages. Any blank pages are indicated.

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1 **(a)** A student creates sound recordings.

Making changes to a sound recording may impact the accuracy of the resulting sound file or the size of the file.

Draw **two** lines from each change to the impacts it has on the sound file.

Change	Impact
increase the duration of the recording	the file size gets bigger
	no change in the file size
increase the sampling rate	the file size gets smaller
	the accuracy of the sound file improves
decrease the sampling resolution	no change to the accuracy of the sound file
	the accuracy of the sound file worsens

[3]

(b) The file names of the sound recordings are stored using the ASCII character set.

Explain how text is represented by the ASCII character set.

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..... [2]

(c) Give **two** differences between the ASCII and UNICODE character sets.

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[2]

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2 **(a)** Data in a computer system is represented in binary.

Put **one** tick (✓) in each row to identify the minimum number of bits used to store each example of data.

Example of data	Number of bits						
	4	**8**	**16**	**24**	**32**	**64**	**128**
the hexadecimal value F139							
16 000 000 unique amplitude values							
an IPv4 address							
256 unique colours							
an IPv6 address							
the denary value 65 000							

[3]

(b) Convert the denary number –108 into a 12-bit two's complement binary number.

..... [1]

(c) A three-place arithmetic shift to the right is performed on the following two's complement negative integer.

Show the result of this arithmetic shift.

10010011

..... [1]

(d) Convert the following positive binary integer into hexadecimal.

1110001100111011

..... [1]

3 **(a)** Complete the truth table for the following logic expression:

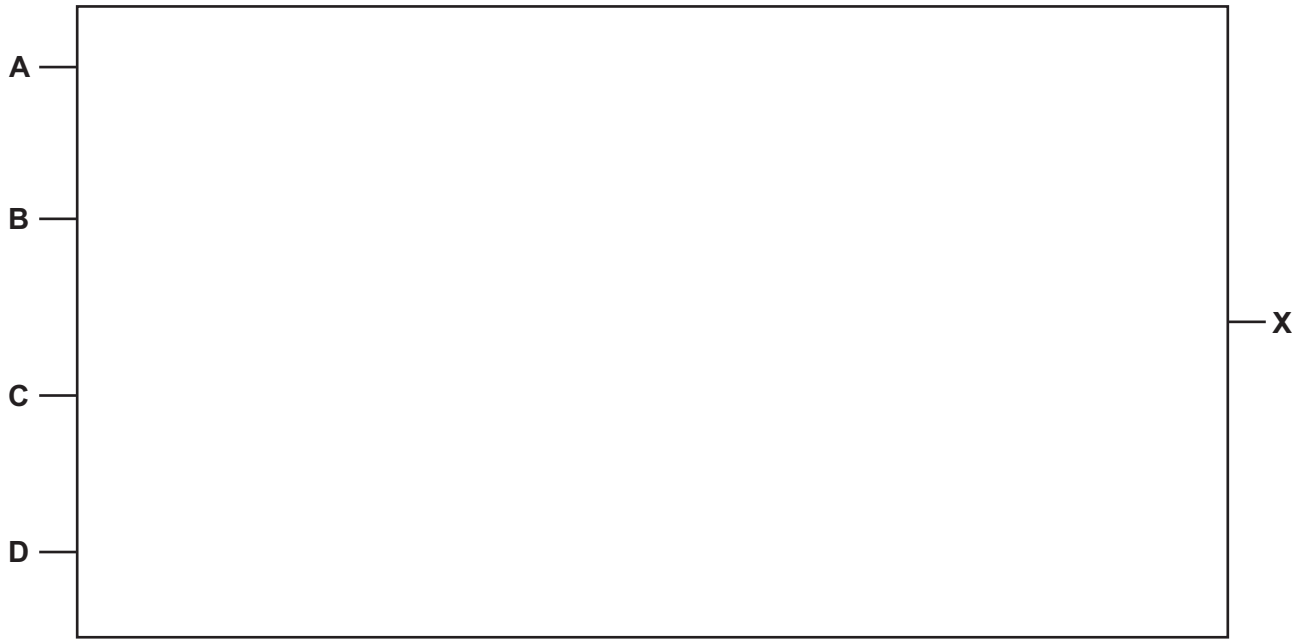
X = (**A** NOR **B**) NAND (**C** XOR **B**)

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw a logic circuit for the logic expression:

X = NOT (((**A** AND **B**) OR **C**) AND (**C** NOR **D**))



[2]

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4 A network engineer is setting up a Local Area Network (LAN) in an office. The network uses both wired and wireless connectivity.

(a) Give **three** advantages of a wireless network compared to a wired network.

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[3]

(b) Give **one** drawback of using a wireless network.

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..... [1]

(c) Complete the table by writing the name of the type of memory technology for each description.

Description	Memory technology
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	
A read only memory where the contents can be written only once after manufacture.	

[3]

5 A company creates a relational database to store data about its customers.

The database, REVIEWS, stores data about the customers, products, complaints and staff.

The database has four tables:

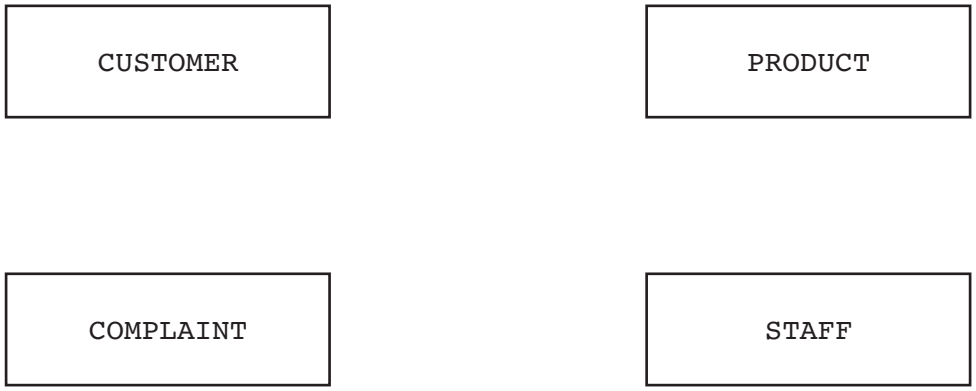
CUSTOMER(CustomerID, CustomerFirstName, CustomerLastName, CustomerEmail)

PRODUCT(ProductID, ProductName, ProductDetail, Price, Rating)

COMPLAINT(ComplaintID, ProductID, CustomerID, ComplaintDetails, StaffID)

STAFF(StaffID, StaffFirstName, StaffLastName, Department, RemoteWorker)

(a) Complete the entity-relationship (E-R) diagram for the database REVIEWS.



[3]

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(b) Some example data from the `STAFF` table is shown.

StaffID	StaffFirstName	StaffLastName	Department	RemoteWorker
1	Ralph	Jura	E	Yes
2	Luca	Emcee	A	Yes
3	Darwin	Acula	F	No

Write a Structured Query Language (SQL) script to define the table `STAFF`.

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..... [4]

(c) Products are given a rating between 1 and 10 inclusive.

Write an SQL script to return only the `ProductID`, `ProductName` and `ComplaintDetails` for all products with a rating of 5 or less. The results need to be displayed in descending order of rating.

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(d) A Database Management System (DBMS) is used to maintain and manage the database.

Describe **two** ways in which the DBMS can be used to ensure the security of the customer data.

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Question 6 starts on the next page.

6 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	**Operand**	
LDM	#n	Immediate addressing. Load the number n to the ACC
AND	#n / Bn / &n	Bitwise AND operation of the contents of the ACC with the operand
AND	<address>	Bitwise AND operation of the contents of the ACC with the contents of <address>
XOR	#n / Bn / &n	Bitwise XOR operation of the contents of the ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of the ACC with the contents of <address>
OR	#n / Bn / &n	Bitwise OR operation of the contents of the ACC with the operand
OR	<address>	Bitwise OR operation of the contents of the ACC with the contents of <address>
CMP	#n	Compare the contents of the ACC with number n
CMP	<address>	Compare the contents of the ACC with the contents of <address>
LSL	#n	Bits in the ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in the ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #127 B denotes a binary number, e.g. B10010001 & denotes a hexadecimal number, e.g. &4A		

The current contents of main memory are shown:

Address	Data
100	0000 0011
101	1010 1110
102	1100 1100
103	1111 1111
104	1100 1100

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(a) Complete the table by writing the contents of the ACC after the execution of each instruction.

Current contents of the ACC	Instruction	Contents of the ACC after the execution of the instruction
0000 1111	AND 101	
0000 0000	LDM #100	
0000 0001	XOR &F1	
0001 0001	CMP 101	

[4]

(b) The Von Neumann model for a computer system uses registers.

Describe the role of the Memory Address Register (MAR) and Memory Data Register (MDR) in the fetch-execute (F-E) cycle.

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..... [4]

(c) Assembly language instructions are grouped.

Complete each statement by writing the name of the appropriate instruction group.

Loading data into the accumulator is an example of an instruction in the

..... group. Incrementing the index register is an example

of an instruction in the ..... group. Branching to another

address is an example of an instruction in the ..... group.

[3]

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7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(a) Describe **one** ethical impact of this use of AI in the classroom.

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..... [2]

(b) A video is made of many bitmap images called frames. 30 frames are recorded every second.

Each frame is 4000 pixels wide by 3000 pixels high. The video records using 16-bit colour depth.

Calculate an estimate for the file size for **one** second of the video in gigabytes.

Show your working.

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File size ..... gigabytes

[2]

(c) Identify **and** describe **one** method of data verification that can be used when transferring data from the digital video camera to the server.

Method .....

Description .....

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[3]

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(d) Describe **two** reasons why the server that stores the videos uses magnetic hard disks instead of solid state (flash) memory.

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(e) The table contains three algorithms that perform data validation. The third algorithm uses the functions `MID` and `LENGTH`.

The functions `MID` and `LENGTH` are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER) RETURNS STRING` returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING) RETURNS INTEGER` returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation

`INPUT x`
 	
`IF x < 1 or x > 26 THEN`
 	
`  OUTPUT "Invalid"`
 	
`ENDIF`	

`INPUT x`
 	
`IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN`
 	
`  OUTPUT "Invalid"`
 	
`ENDIF`	

`INPUT x`
 	
`FLAG ← FALSE`
 	
`FOR INDEX ← 1 TO LENGTH(x)`
 	
`  IF MID(x, INDEX, 1) = "@" THEN`
 	
`  FLAG ← TRUE`
 	
`  ENDIF`
 	
`NEXT INDEX`
 	
`IF NOT FLAG THEN`
 	
`  OUTPUT "Invalid"`
 	
`ENDIF`	

[3]

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8 Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(a) State **two** disadvantages of using a compiler compared to an interpreter during program development.

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[2]

(b) Explain how a programmer benefits from using program libraries during program development.

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..... [3]

(c) A programmer is developing an Operating System (OS).

Identify **two** types of software licence the programmer can use to allow other people to edit and redistribute the OS.

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2 .....

[2]

(d) Memory management is one key management task performed by an OS.

Give **three** other key management tasks that are performed by an OS.

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